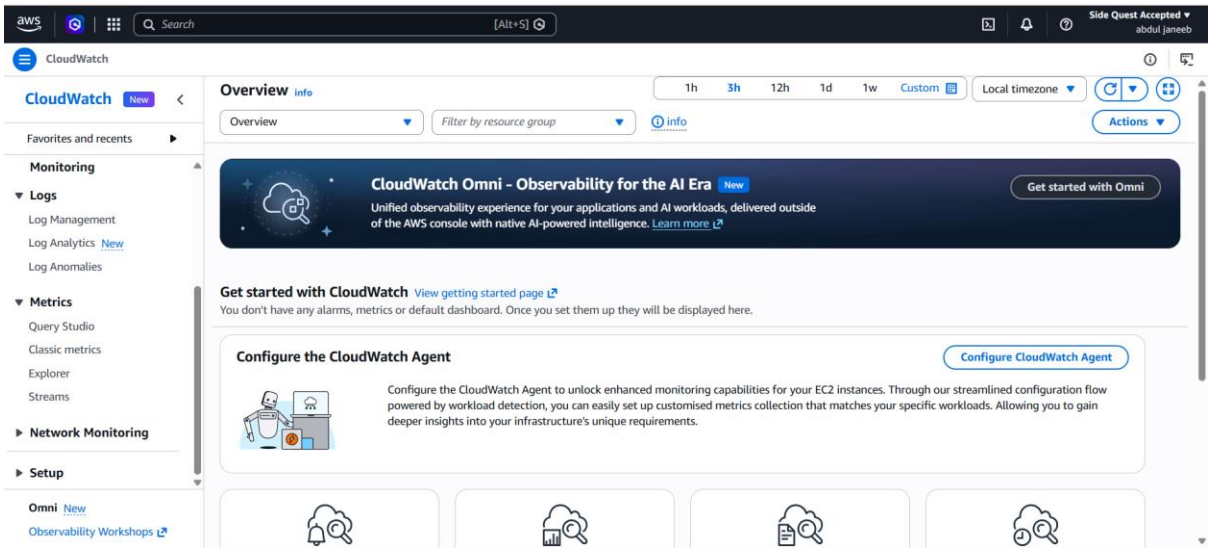
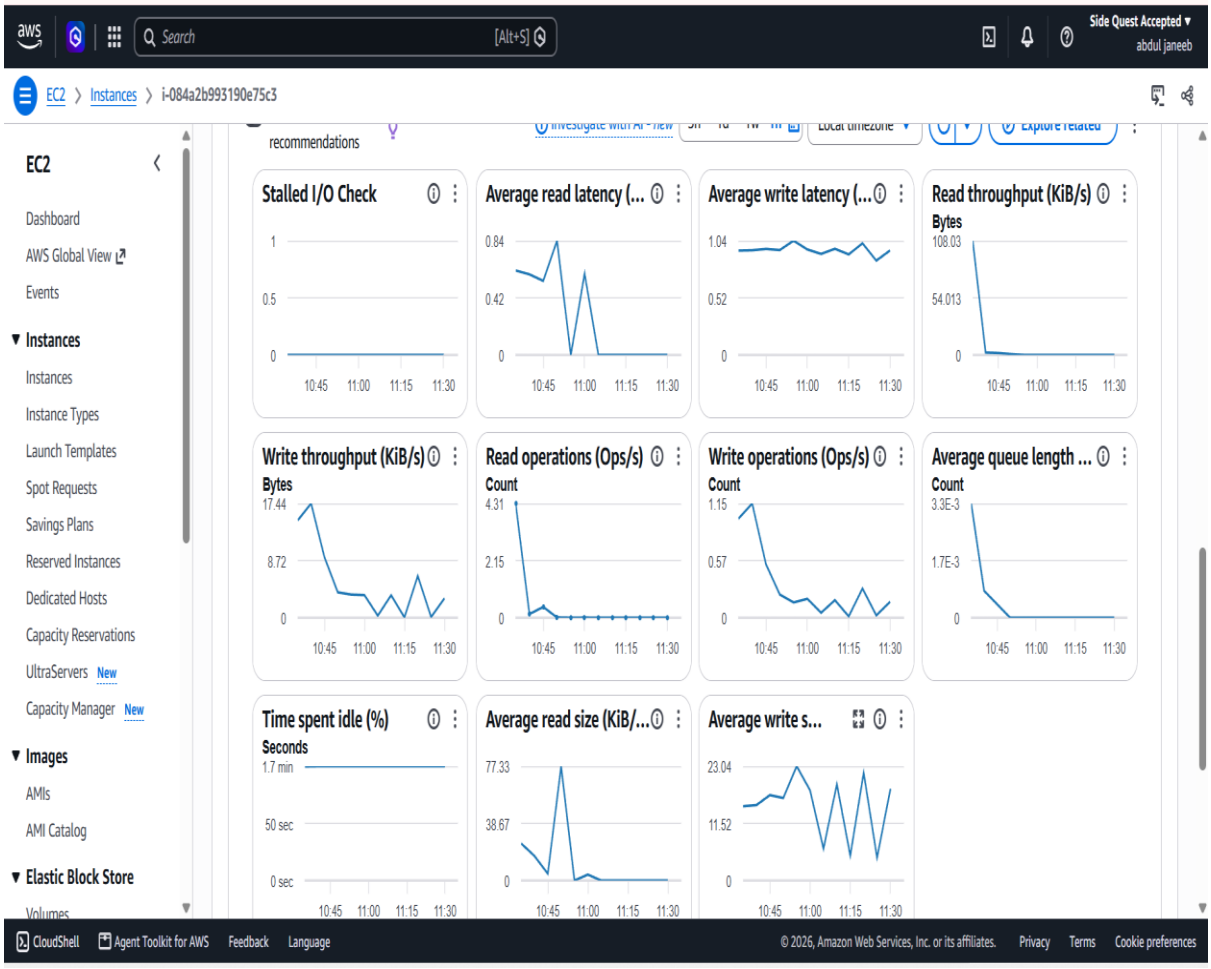
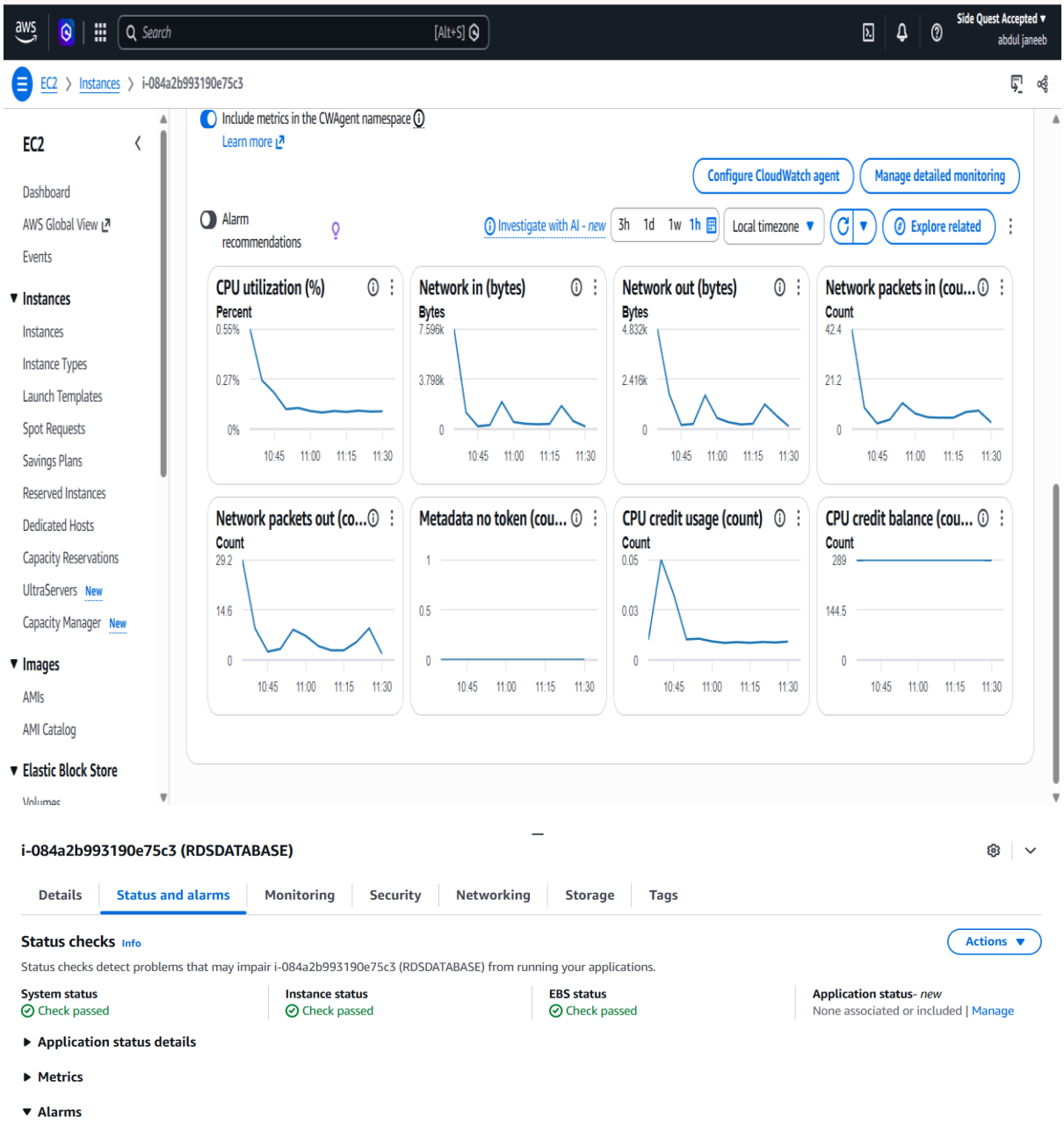
AWS CloudWatch Hands-on Lab Tasks

- **Explored the Cloudwatch Console**



- **Task 2: Monitor an EC2 Instance**

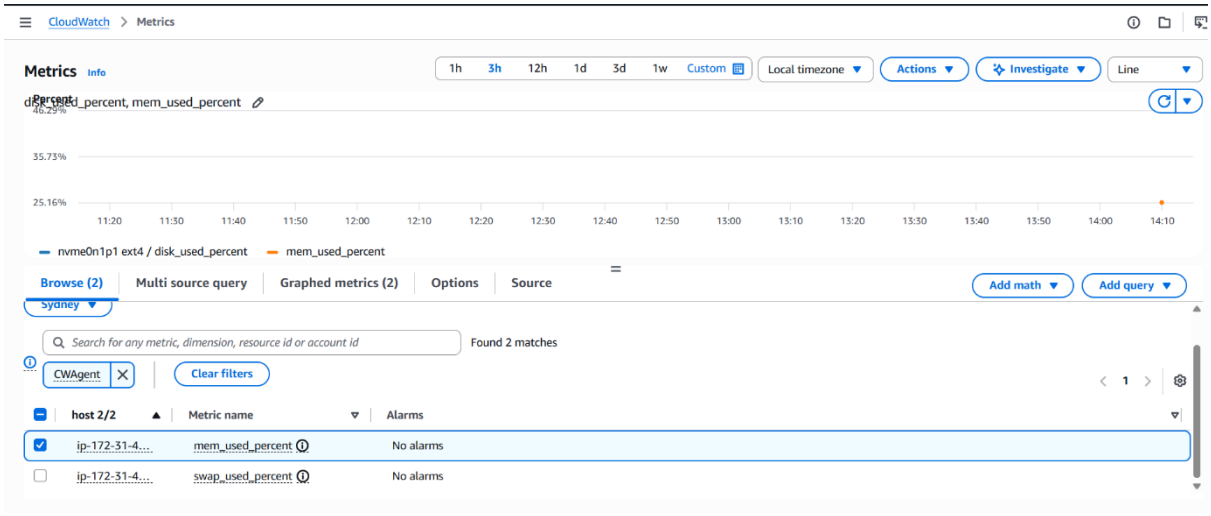


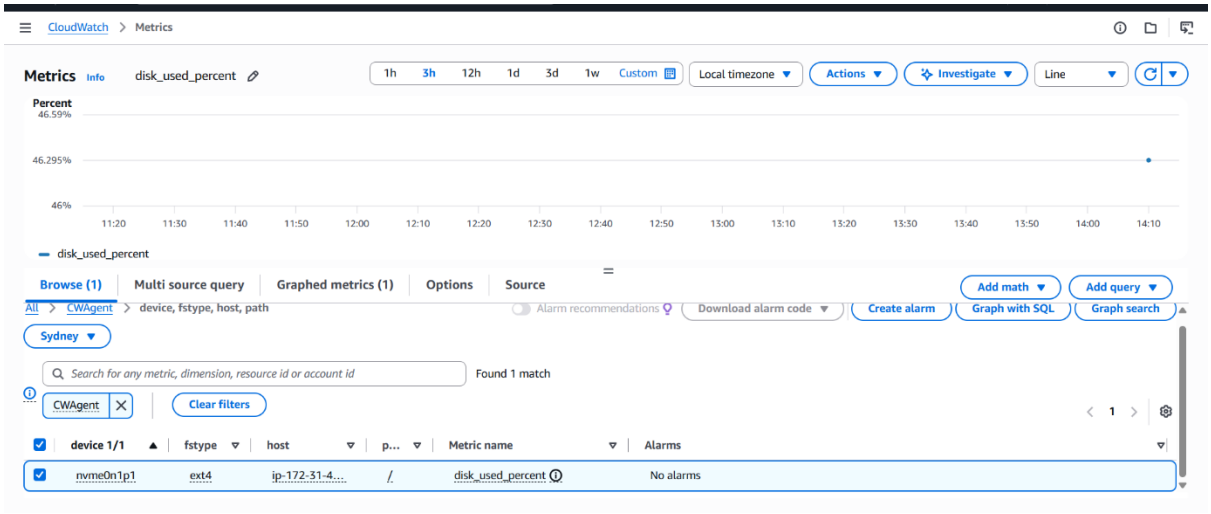


- Task 3: Install the CloudWatch Agent

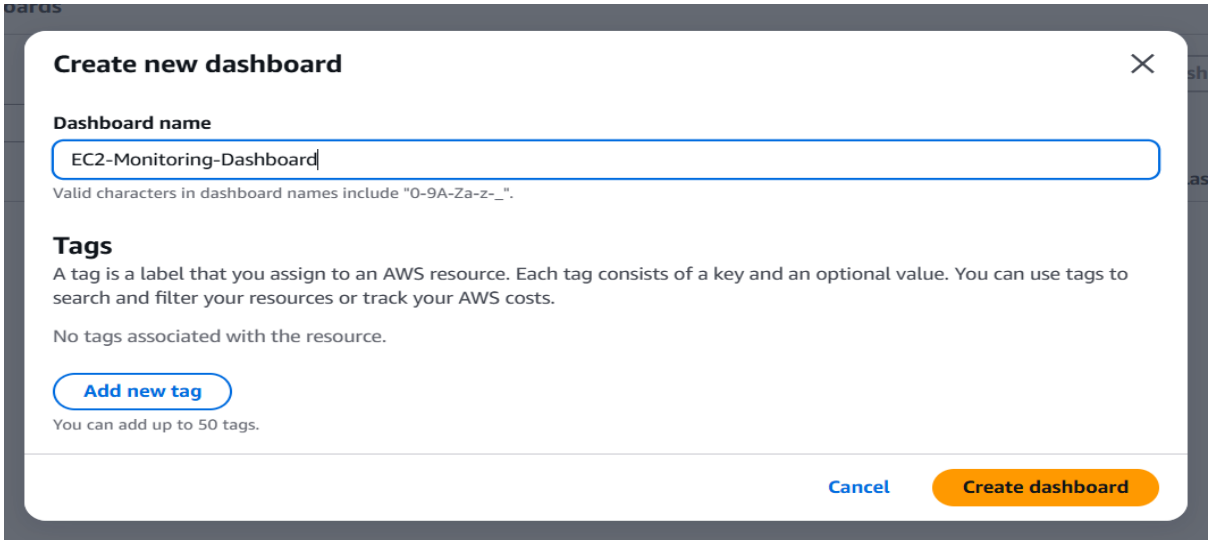
- Connected to the Ubuntu EC2 instance through SSH and checked the existing RAM, disk, and swap usage using `free -h`, `df -h`, and `swapon --show`. Updated the package information using `sudo apt update`, downloaded the Amazon CloudWatch Agent package using `wget`, and installed it using `dpkg`. Created a JSON configuration file to collect RAM utilization (`mem_used_percent`), swap utilization (`swap_used_percent`), and disk utilization (`used_percent`) at 60-second intervals under the `CWAgent` namespace. Created an IAM role with the `CloudWatchAgentServerPolicy` permission and attached it to the EC2 instance to allow the agent to publish metrics. Loaded the configuration and started the CloudWatch Agent using the `amazon-cloudwatch-agent-ctl` command. Verified that the agent was running and navigated to CloudWatch → Metrics → All metrics →

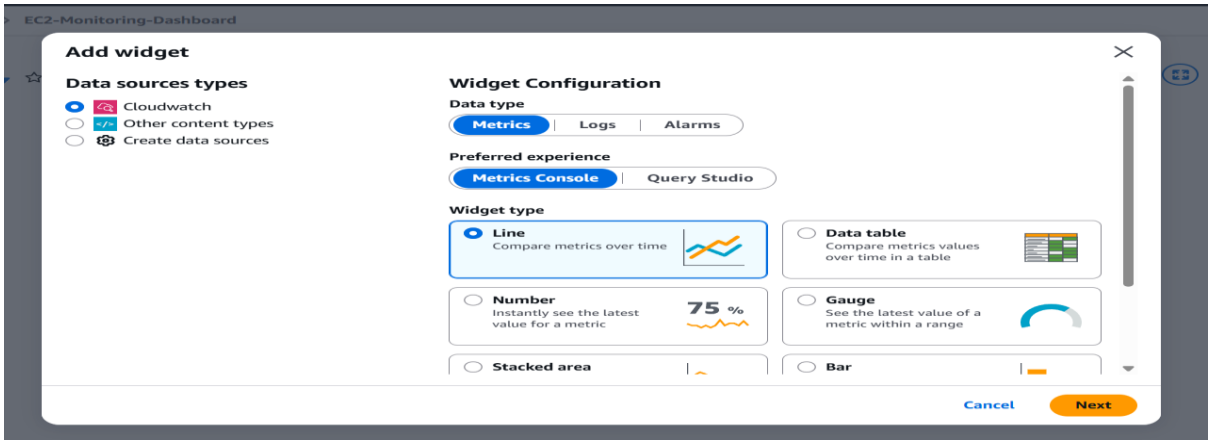
CWAgent to view the RAM, Swap, and Disk utilization metrics. Successfully configured and verified EC2 system-level monitoring using Amazon CloudWatch.

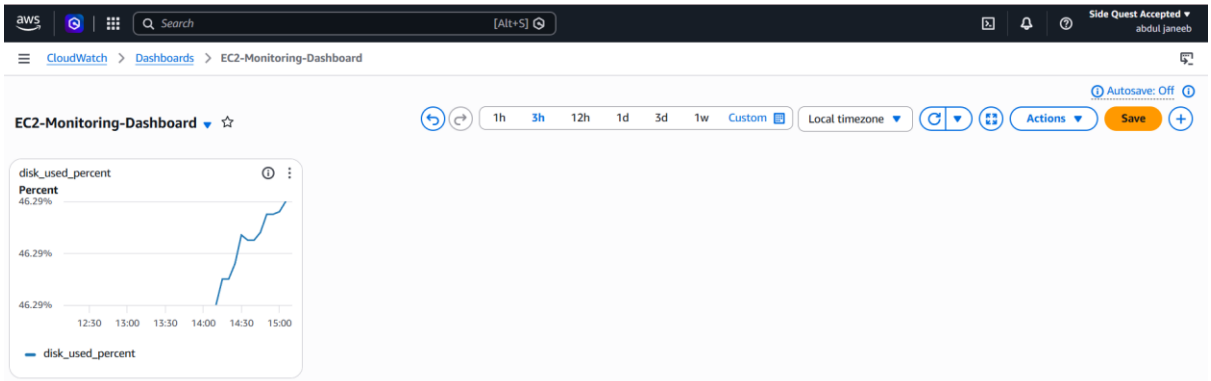


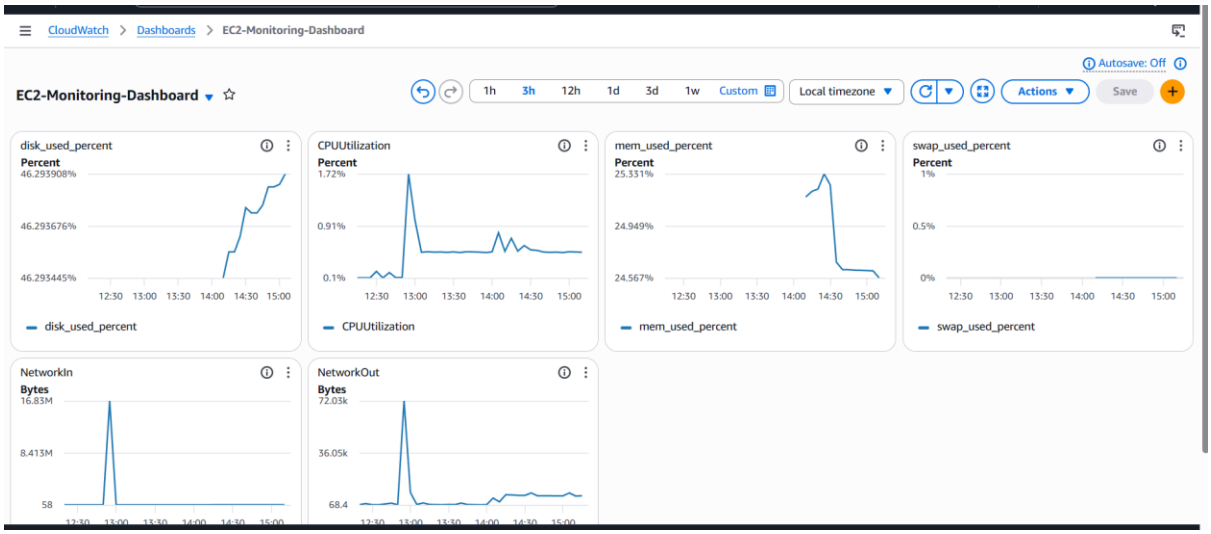


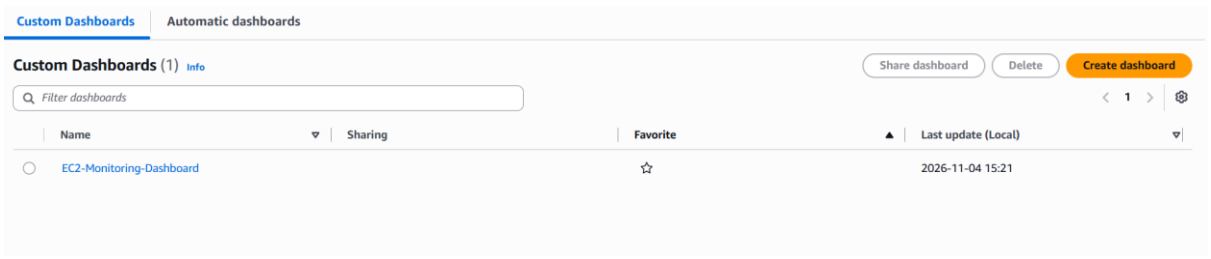
Task 4: Create a CloudWatch Dashboard



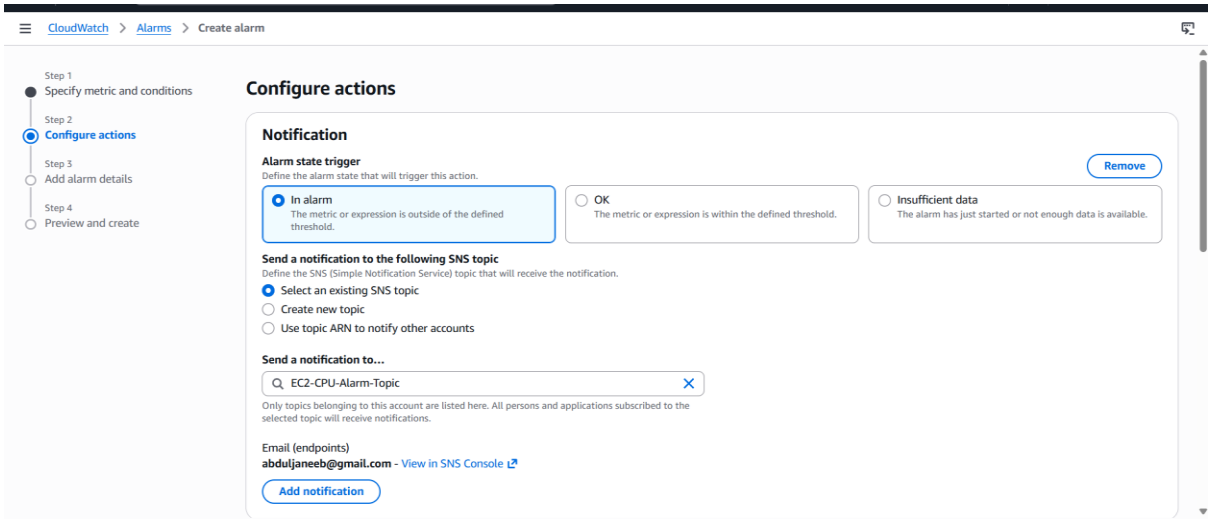

EC2-Monitoring-Dashboard
Add widget
Data sources types
Cloudwatch
Other content types
Create data sources
Widget Configuration
Data type
Metrics
Logs
Alarms
Preferred experience
Metrics Console
Query Studio
Widget type
Line
Compare metrics over time
Data table
Compare metrics values over time in a table
Number
Instantly see the latest value for a metric
75 %
Gauge
See the latest value of a metric within a range
Stacked area
Bar
Cancel
Next

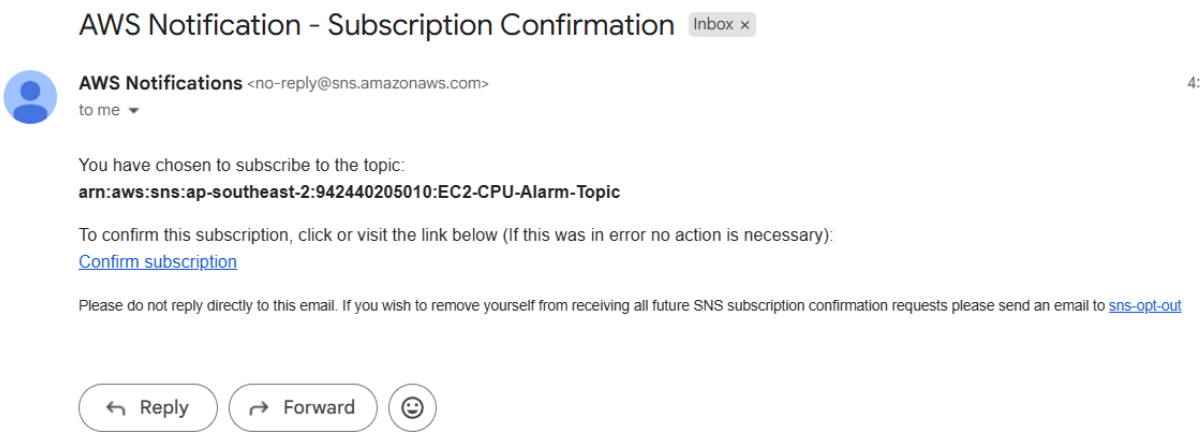

Search
[Alt+S]
Side Quest Accepted
abdul janeeb
CloudWatch
Dashboards
EC2-Monitoring-Dashboard
Autosave: Off
EC2-Monitoring-Dashboard
1h
3h
12h
1d
3d
1w
Custom
Local timezone
Actions
Save
disk_used_percent
Percent
46.29%
46.29%
46.29%
12:30
13:00
13:30
14:00
14:30
15:00
disk_used_percent

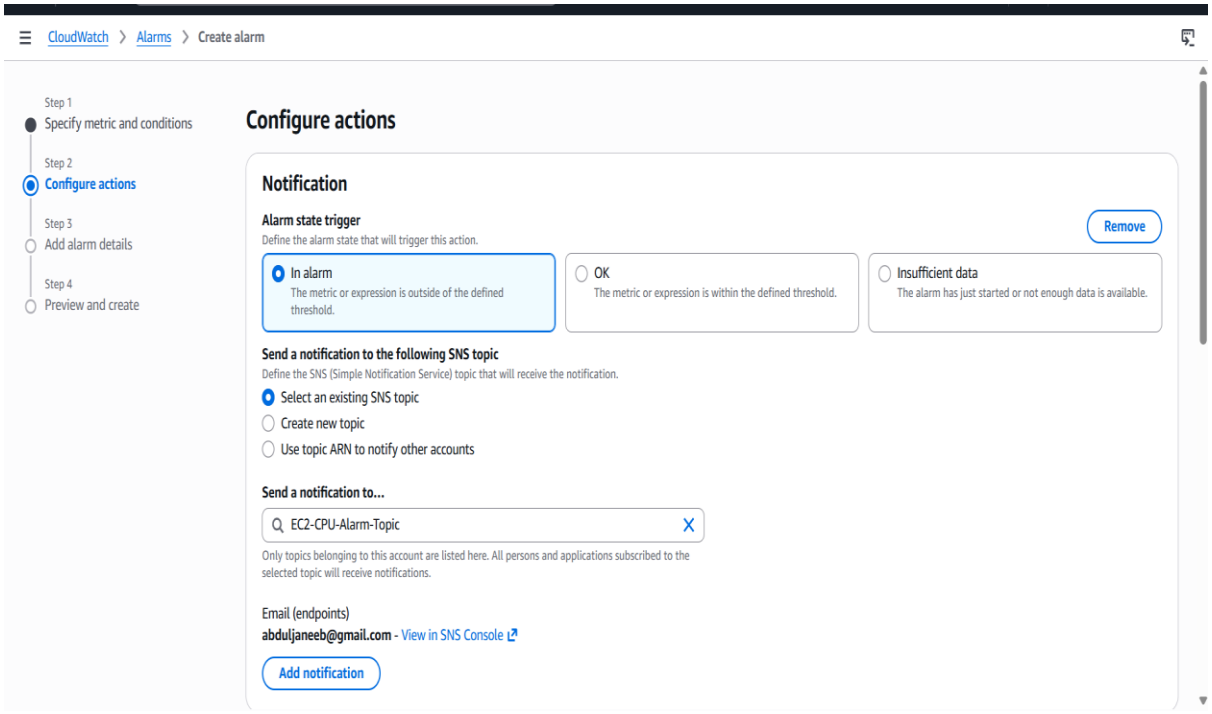

CloudWatch
Dashboards
EC2-Monitoring-Dashboard
Autosave: Off
EC2-Monitoring-Dashboard
1h
3h
12h
1d
3d
1w
Custom
Local timezone
Actions
Save
disk_used_percent
Percent
46.293908%
46.293676%
46.293445%
disk_used_percent
CPUUtilization
Percent
1.72%
0.91%
0.1%
CPUUtilization
mem_used_percent
Percent
25.331%
24.949%
24.567%
mem_used_percent
swap_used_percent
Percent
1%
0.5%
0%
swap_used_percent
NetworkIn
Bytes
16.83M
8.413M
58
NetworkOut
Bytes
72.03k
36.05k
68.4
12:30
13:00
13:30
14:00
14:30
15:00


Custom Dashboards
Automatic dashboards
Custom Dashboards (1)
Info
Share dashboard
Delete
Create dashboard
Filter dashboards
1
Name
Sharing
Favorite
Last update (Local)
EC2-Monitoring-Dashboard
2026-11-04 15:21

Task 5: Create a CloudWatch Alarm









Task 6: Generate High CPU Utilization

```
Last login: Wed Nov  4 06:37:27 2026 from 13.239.158.3
ubuntu@ip-172-31-47-82:~$ sudo apt update
sudo apt install stress -y
Hit:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
27 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  stress

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 27
  Download size: 18.2 kB
  Space needed: 52.2 kB / 3801 MB available

Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 stress amd64 1.0.7-2 [18.2 kB]
Fetched 18.2 kB in 0s (917 kB/s)
Selecting previously unselected package stress.
(Reading database ... 124338 files and directories currently installed.)
Preparing to unpack .../stress_1.0.7-2_amd64v3.deb ...
Unpacking stress (1.0.7-2) ...
Setting up stress (1.0.7-2) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  7.0.0-1014-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 7.0.0-1016-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.
```

```
Diagnostics:
  The currently running kernel version is not the expected kernel version 7.0.0-1016-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

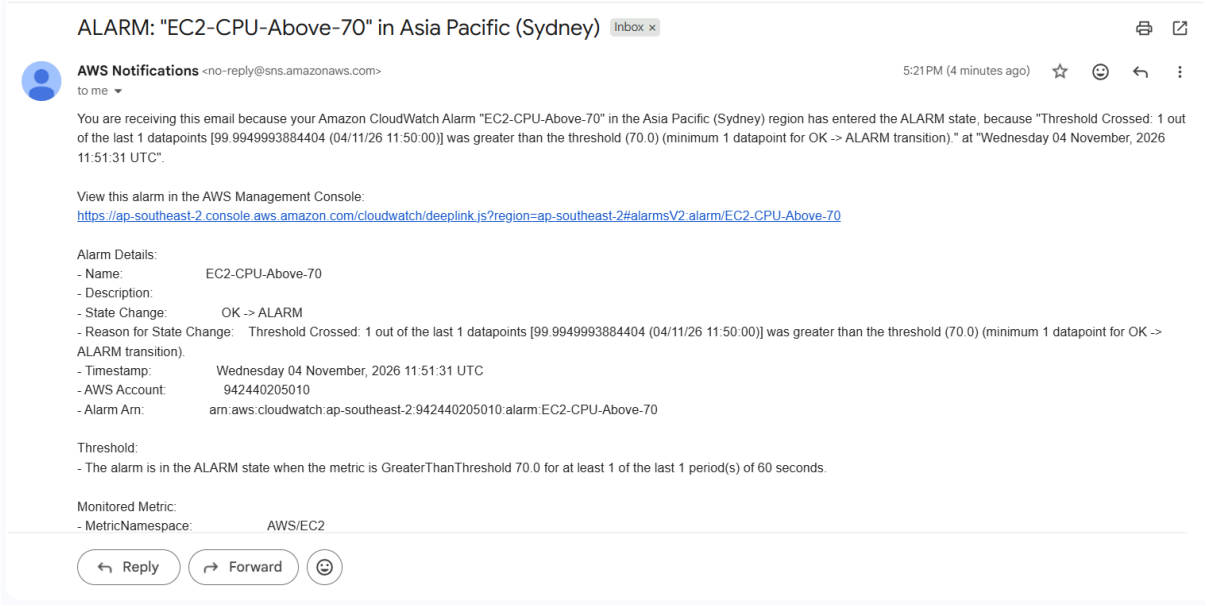
Restarting services...

Service restarts being deferred:
 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-47-82:~$ stress --cpu 2 --timeout 300
stress: info: [22887] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
```



ALARM: "EC2-CPU-Above-70" in Asia Pacific (Sydney) Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 5:21 PM (4 minutes ago)
to me

You are receiving this email because your Amazon CloudWatch Alarm "EC2-CPU-Above-70" in the Asia Pacific (Sydney) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [99.9949993884404 (04/11/26 11:50:00)] was greater than the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Wednesday 04 November, 2026 11:51:31 UTC".

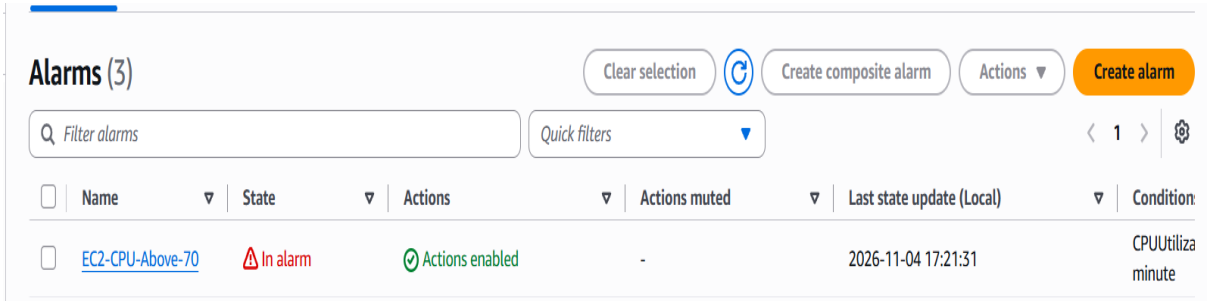
View this alarm in the AWS Management Console:
https://ap-southeast-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=ap-southeast-2#alarmsV2:alarm/EC2-CPU-Above-70

Alarm Details:
- Name: EC2-CPU-Above-70
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [99.9949993884404 (04/11/26 11:50:00)] was greater than the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Wednesday 04 November, 2026 11:51:31 UTC
- AWS Account: 942440205010
- Alarm Arn: arn:aws:cloudwatch:ap-southeast-2:942440205010:alarm:EC2-CPU-Above-70

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanThreshold 70.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:
- MetricNamespace: AWS/EC2

Reply Forward

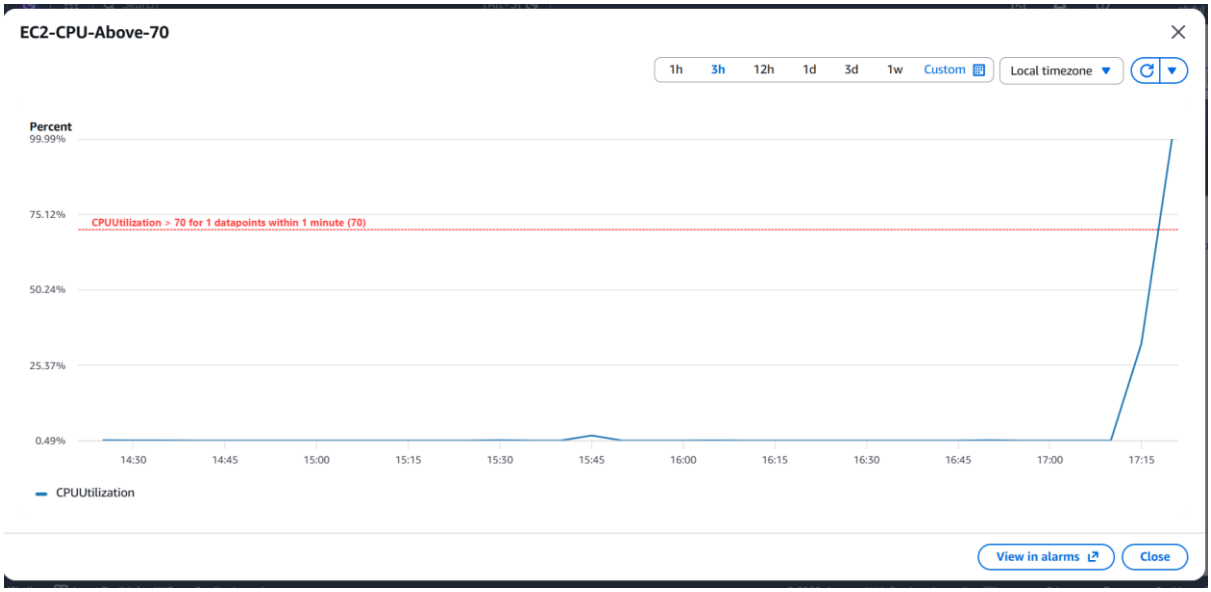


Alarms (3)

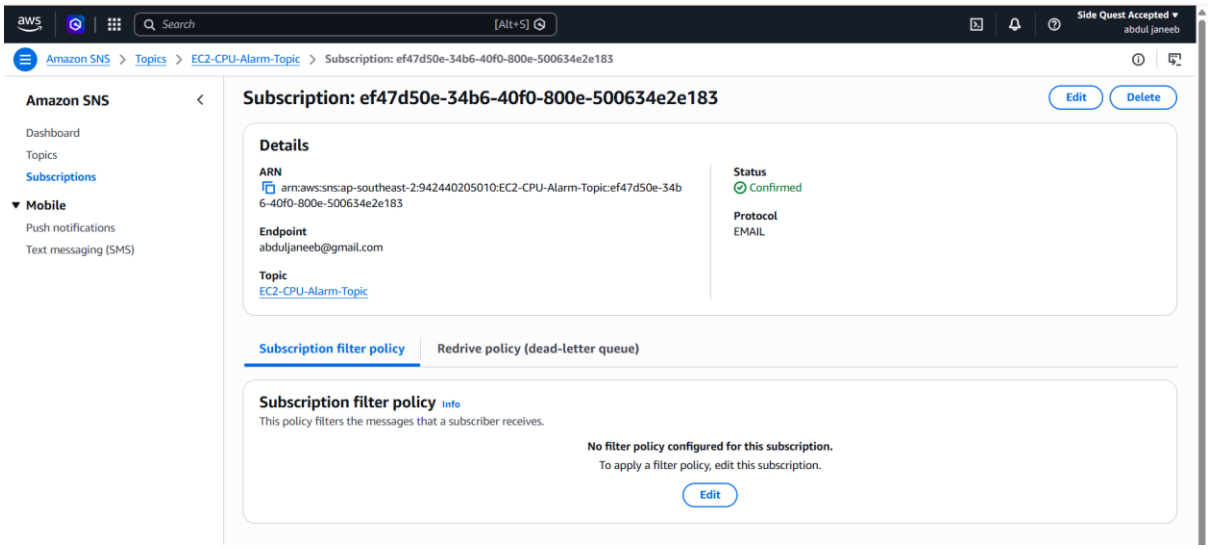
Clear selection | Create composite alarm | Actions | Create alarm

Filter alarms | Quick filters | 1

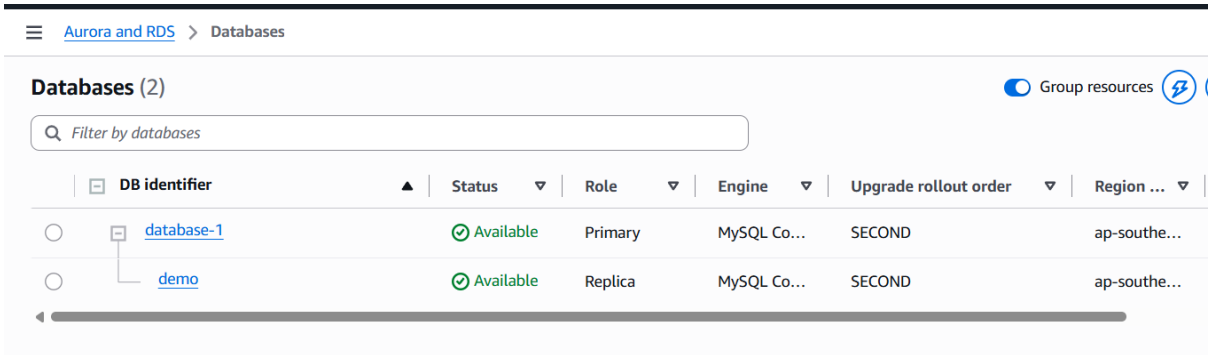
Name	State	Actions	Actions muted	Last state update (Local)	Condition
EC2-CPU-Above-70	In alarm	Actions enabled	-	2026-11-04 17:21:31	CPUUtiliza minute



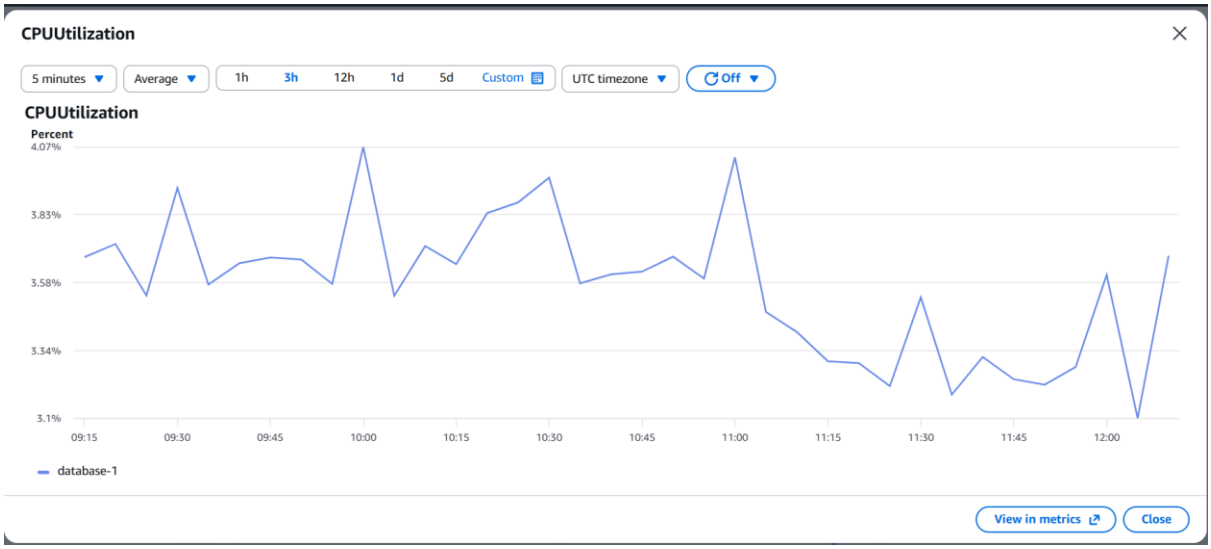
Task 7: Configure Amazon SNS Notifications



Task 8: Monitor an Amazon RDS Database



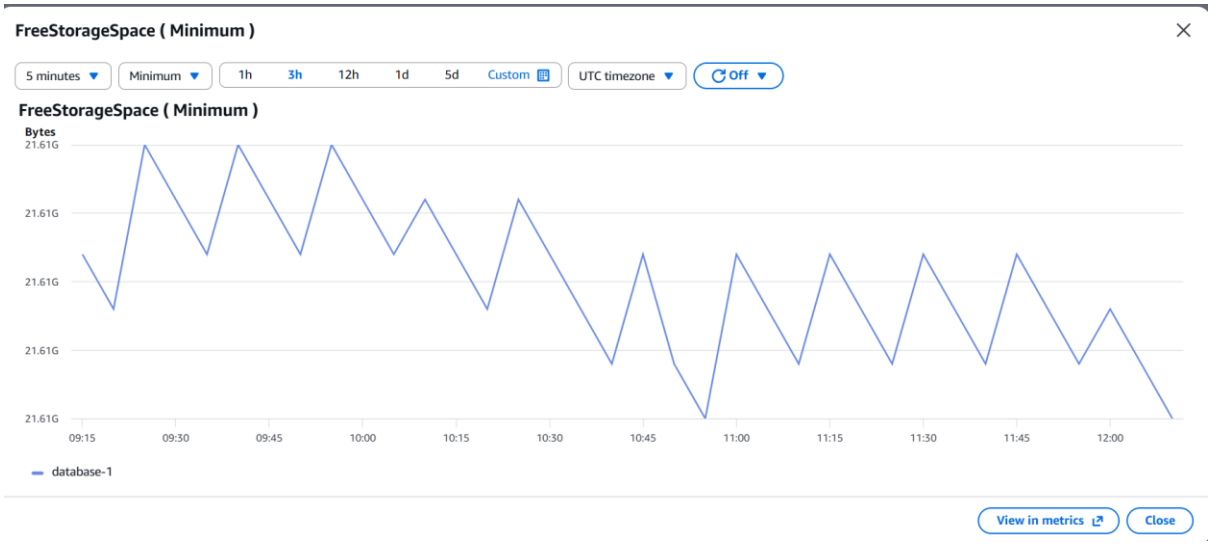
- **CPUUtilization**



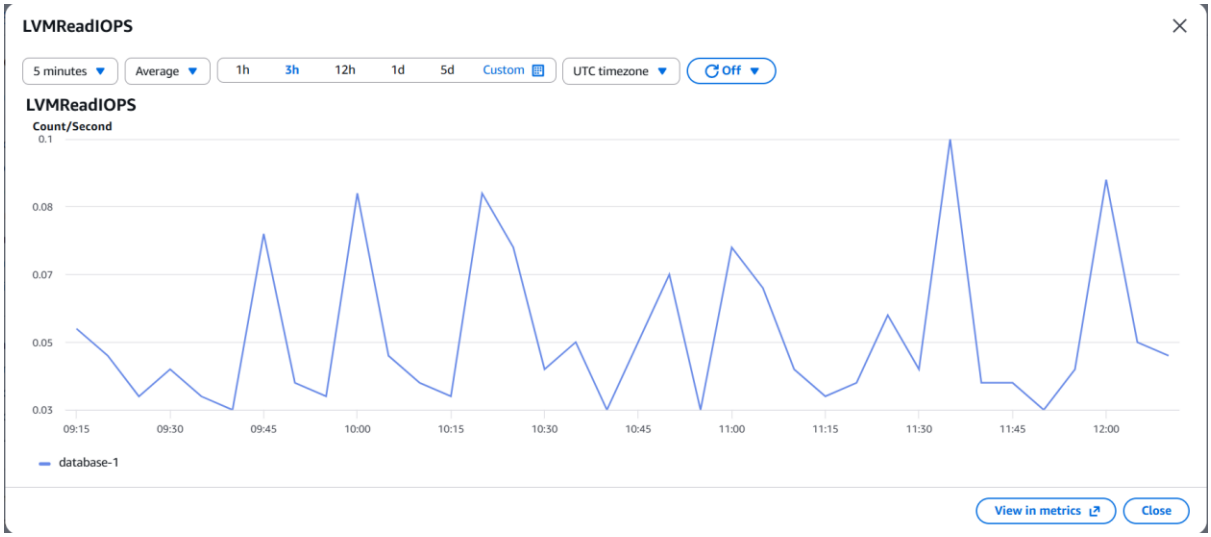
- **DatabaseConnections**



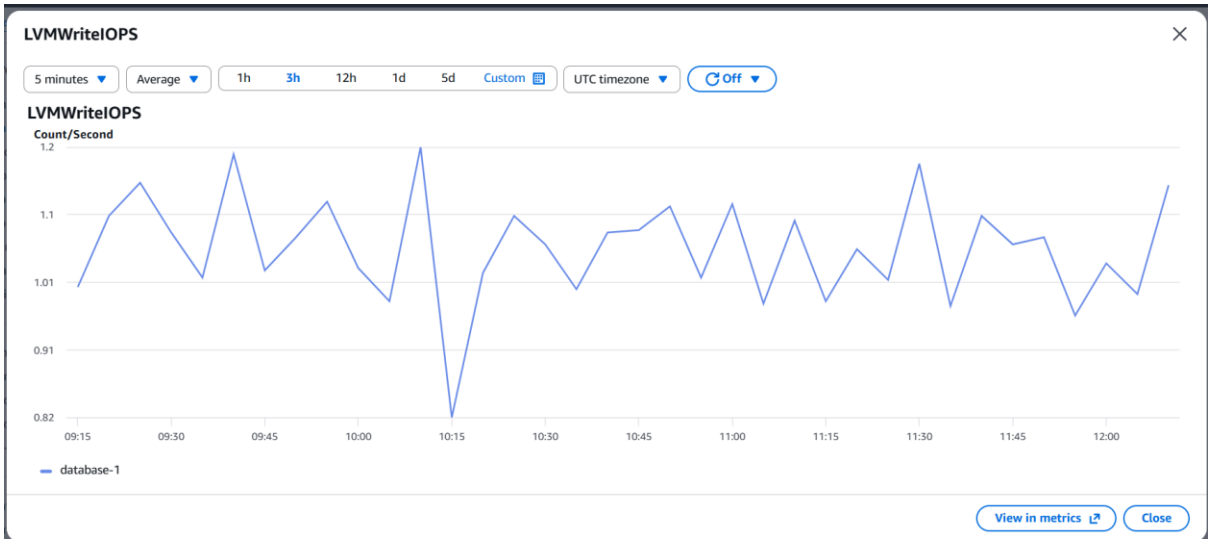
- **FreeStorageSpace**



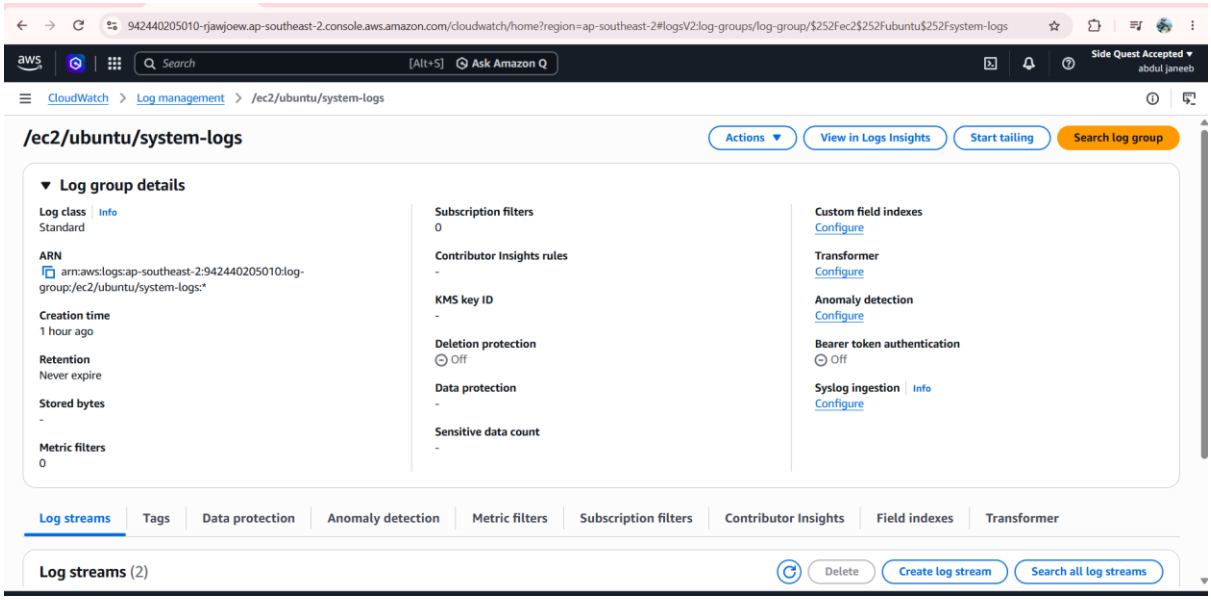
- **LVMReadIOPS**

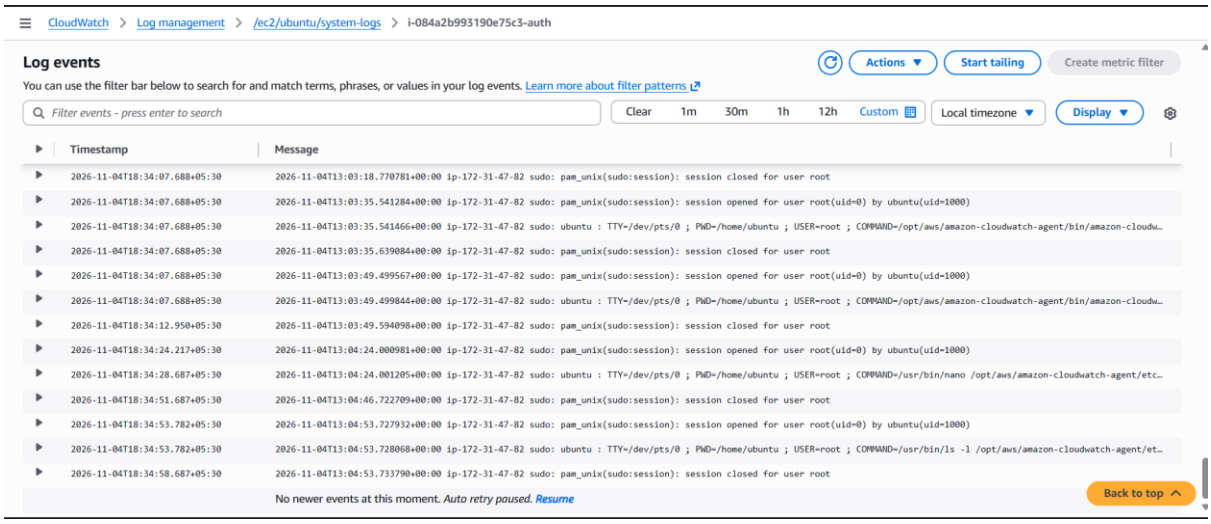


- **LVMWriteIOPS**

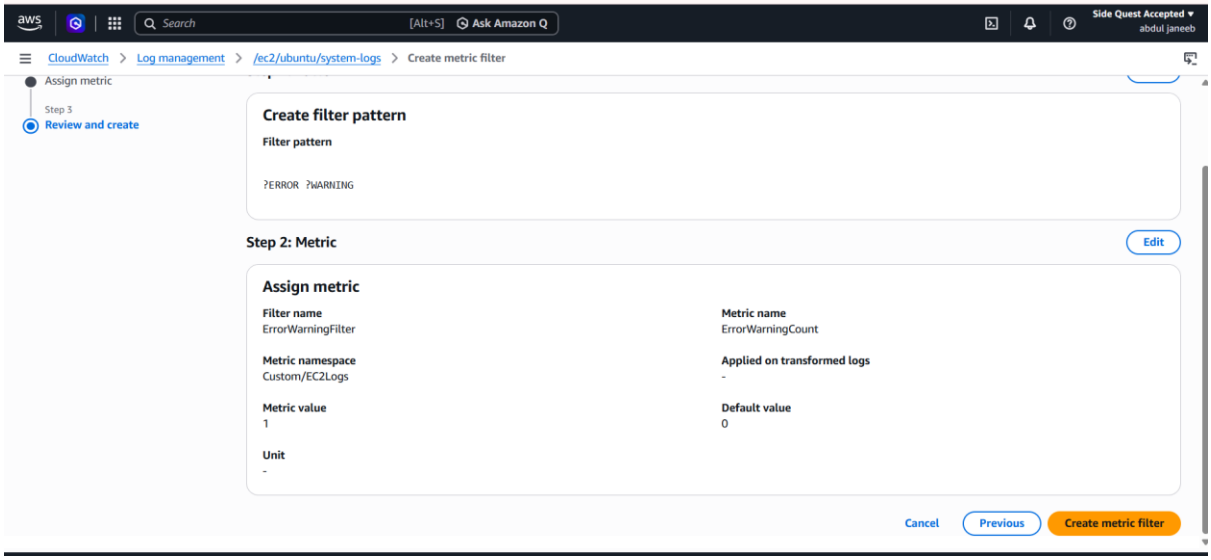


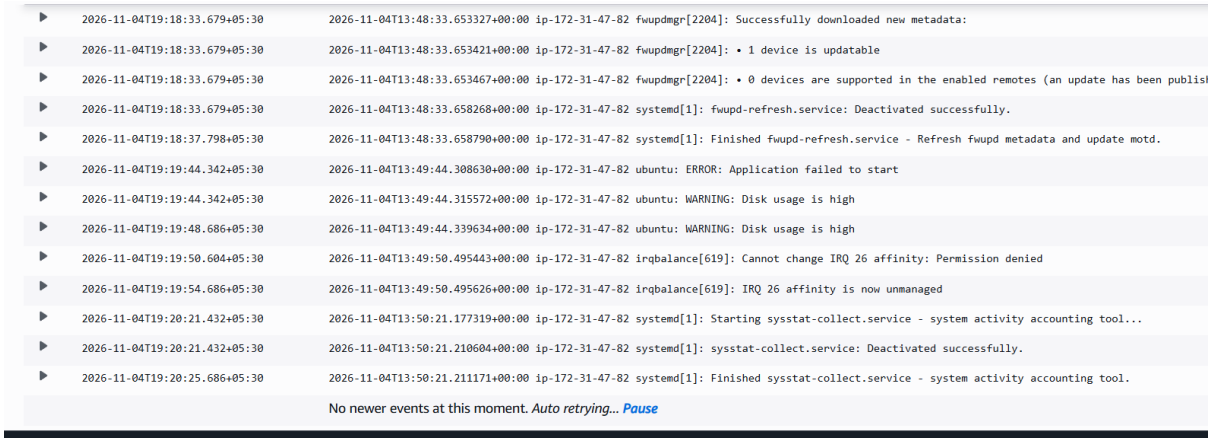
Task 9: Explore CloudWatch Logs

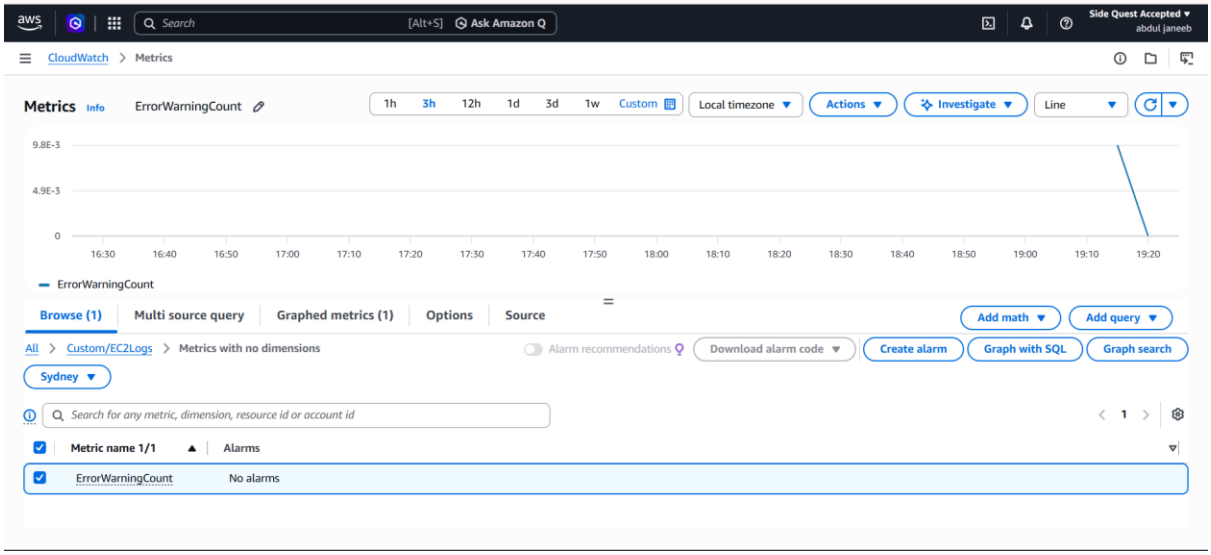




Task 10: Create a Metric Filter

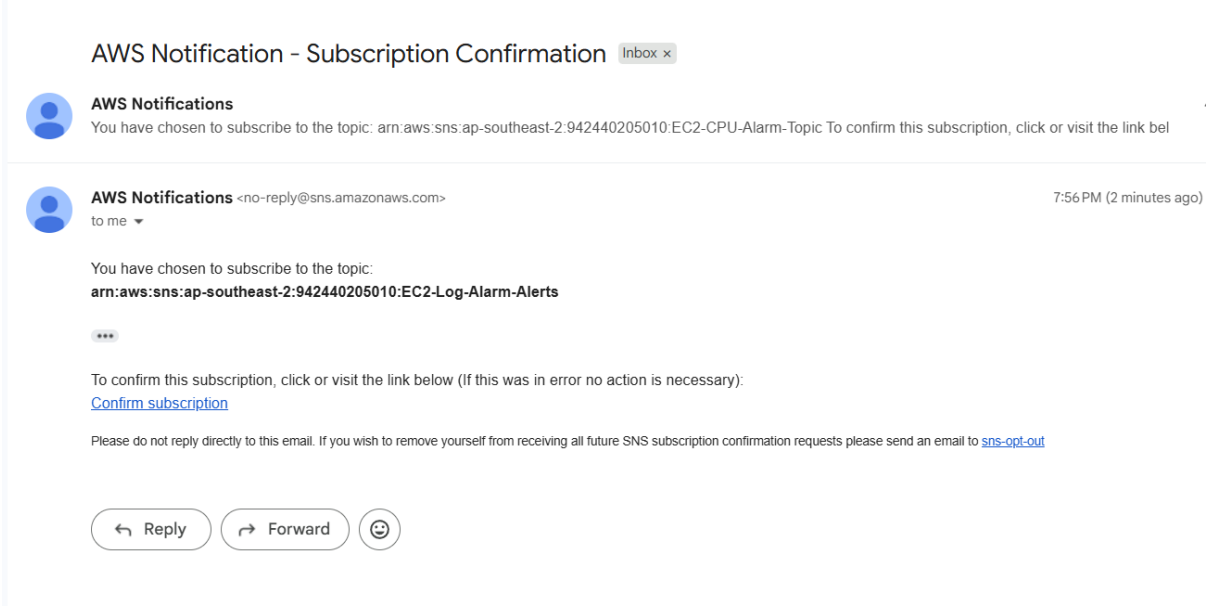


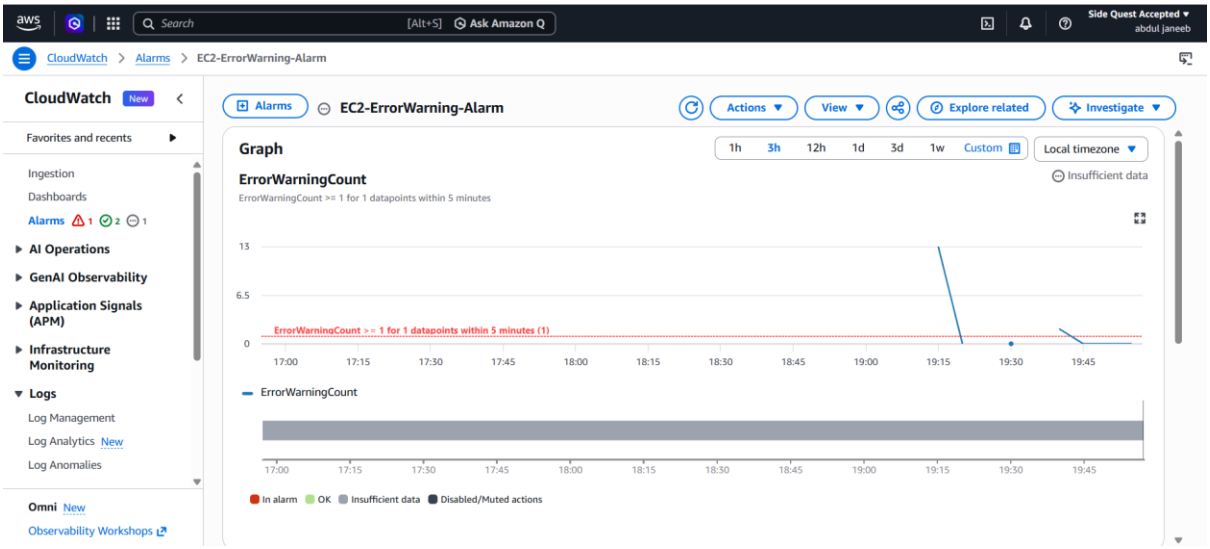




Task 11: Create a CloudWatch Alarm from a Log Metric

Created a CloudWatch alarm named EC2-ErrorWarning-Alarm using the custom metric ErrorWarningCount. Configured the alarm with the Sum statistic, a 5-minute evaluation period, and a threshold of greater than or equal to 1. Configured an SNS notification topic to receive email alerts when ERROR or WARNING log events trigger the alarm. Successfully created the alarm and verified its configuration in the CloudWatch console.

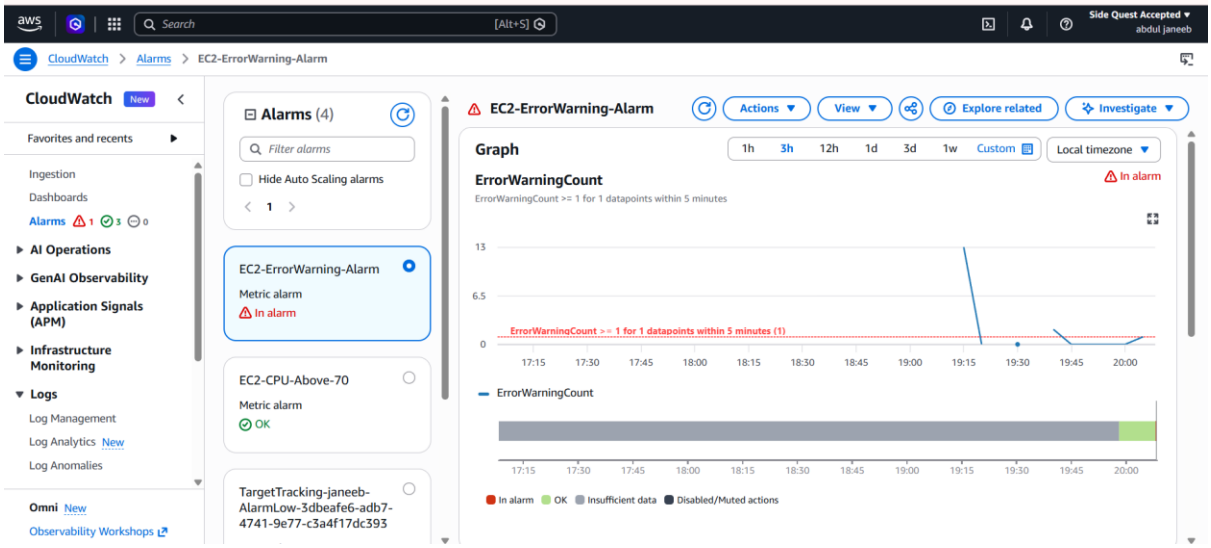




- Testing the threshold using a error log

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Wed Nov  4 13:33:25 2026 from 13.239.158.4
ubuntu@ip-172-31-47-82:~$ logger "ERROR: Testing CloudWatch alarm"
ubuntu@ip-172-31-47-82:~$
```



ALARM: "EC2-ErrorWarning-Alarm" in Asia Pacific (Sydney) Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 8:08 PM (2 minutes ago)

to me

You are receiving this email because your Amazon CloudWatch Alarm "EC2-ErrorWarning-Alarm" in the Asia Pacific (Sydney) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (04/11/26 14:33:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Wednesday 04 November, 2026 14:38:10 UTC".

View this alarm in the AWS Management Console:
https://ap-southeast-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=ap-southeast-2#alarmsV2:alarm/EC2-ErrorWarning-Alarm

Alarm Details:
- Name: EC2-ErrorWarning-Alarm
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (04/11/26 14:33:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Wednesday 04 November, 2026 14:38:10 UTC
- AWS Account: 942440205010
- Alarm Arn: arn:aws:cloudwatch:ap-southeast-2:942440205010:alarm:EC2-ErrorWarning-Alarm

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

Reply Forward

Task 12: Create an AWS Billing Alarm

